

.

EE1060: Differential Equations and Transform Techniques

Lecture 19: Fourier Series

Topics :

1. Solution of a differential equation for periodic inputs
 2. Fourier Series
-

Solution of an ODE for periodic inputs and Orthogonality of Functions

$$a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = \underbrace{g(x)}_{\text{input}} \longrightarrow \textcircled{1} \Rightarrow L[y] = g(x)$$

Observations:-

① $\text{Eq } \textcircled{1} \text{ is linear} \Rightarrow \text{if } g(x) = g_1(x) + g_2(x)$ [$g_1(x)$ and $g_2(x)$ are functions for which the sol: $L[y] = g_i(x)$ is easy to compute]

let $y_1: L[y] = g_1(x)$
 $y_2: L[y] = g_2(x)$

response $y: L[y] = g(x) = y_1 + y_2$

② if $g(x)$ is periodic $g(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 x) + b_n \sin(n\omega_0 x)]$

(Fourier Series)

$\omega_0 = \frac{2\pi}{L} \leftarrow \text{Period of } g(x)$

Trigonometric Fourier Series

then responses:

$y_0: L[y] = a_0$

$y_{c,n}: L[y] = \cos(n\omega_0 x)$

$y_{s,n}: L[y] = \sin(n\omega_0 x)$

$$\left. \begin{array}{l} y_0: L[y] = a_0 \\ y_{c,n}: L[y] = \cos(n\omega_0 x) \\ y_{s,n}: L[y] = \sin(n\omega_0 x) \end{array} \right\} y = y_0 + \sum_{n=1}^{\infty} [a_n y_{c,n} + b_n y_{s,n}]$$

Fourier Series - Trigonometric and Complex Exponential forms

$g(t)$ is Periodic if $\exists T \in \mathbb{R} : g(t+T) = g(t) \quad \forall t.$

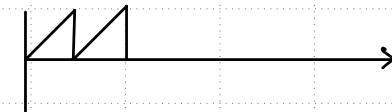
Examples: $\sin(t) \rightarrow T = 2\pi$

$$\left\{ \begin{array}{l} \sin(nt) \rightarrow T = 2\pi/n \\ \cos(nt) \rightarrow T = 2\pi/n \end{array} \right.$$

Square $\rightarrow Cx \quad Dx.$

Triangular $\rightarrow C\sqrt{\quad} \quad Dx$

Sawtooth $\rightarrow Cx \quad Dx.$



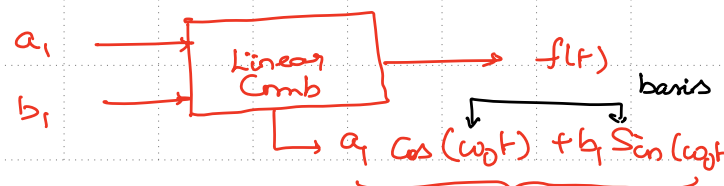
$g(t) =$ summation of sinusoidal & cosinusoidal signal of various freq.

$\downarrow$
Periodic with Period T .

$$\omega_0 = \frac{2\pi}{T}$$

$\rightarrow \sin(\omega_0 t) \text{ \& \& } \cos(\omega_0 t) \rightarrow$ Periodic with period T .

$\downarrow$
must be seen that the $g(t)$ is periodic with same period.



Since every possible $g(t)$ cannot be gen using ①

Complex Exponential Fourier Series - Negative Frequencies

Expand the basis for synthesizing more class of periodic signals.

a) pick terms such that period is T .

$\cos(n\omega_0 t)$ & $\sin(n\omega_0 t)$ for $n = 1, 2, \dots$ ($n=0$ is a special case)
 (fundamental period is T/n yet they are periodic with period T).

Example:-

$$T=2; \quad \omega_0 = \frac{2\pi}{T} = \pi \Rightarrow \text{basis} = \{ \cos(n\pi t), \sin(n\pi t) \} \quad n=0, \dots$$

$$n=0 \rightarrow \text{DC}$$

$$n=1 \rightarrow \text{fundamental}$$

$$n>1 \rightarrow \text{harmonics}$$

given $g(t)$ whose period is T : a) Can it be synthesized as sum of sines & cosines.

$$g(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

b) if so, what will be the coefficients

(a_0, a_n, b_n)

$\rightarrow$ Fourier Coefficients.

Complex Exponential Fourier Series for Real Signals: $C_{-k} = C_k^*$ Negative frequencies.

$$g(t) \text{ as a LC of } \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \quad \text{where } \omega_0 = \frac{2\pi}{T}$$

$\hookrightarrow$ Complex Coefficient

a) negative freq: $\rightarrow$

Need :- To represent real signals.

Inference :- for an object exhibiting uniform circular motion with
freq ω_0

$\omega_0 > 0 \rightarrow$ object moving in counter clockwise direction

$\omega_0 < 0 \rightarrow$ " " " clockwise direction.